# MSG No. 51.289 $Pmma$ [ Type I, orthorhombic ]

Table 1: Wyckoff site: `2a`, site symmetry: `.2/m.`

| No. | position | mapping |
|---|---|---|
| 1 | $[0, 0, 0]$ | `[1,3,5,7]` |
| 2 | $\left[\frac{1}{2}, 0, 0\right]$ | `[2,4,6,8]` |

Table 2: Wyckoff site: `2b`, site symmetry: `.2/m.`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0, \frac{1}{2}, 0\right]$ | `[1,3,5,7]` |
| 2 | $\left[\frac{1}{2}, \frac{1}{2}, 0\right]$ | `[2,4,6,8]` |

Table 3: Wyckoff site: `2c`, site symmetry: `.2/m.`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0, 0, \frac{1}{2}\right]$ | `[1,3,5,7]` |
| 2 | $\left[\frac{1}{2}, 0, \frac{1}{2}\right]$ | `[2,4,6,8]` |

Table 4: Wyckoff site: `2d`, site symmetry: `.2/m.`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0, \frac{1}{2}, \frac{1}{2}\right]$ | `[1,3,5,7]` |
| 2 | $\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$ | `[2,4,6,8]` |

Table 5: Wyckoff site: `2e`, site symmetry: `mm2`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4}, 0, z\right]$ | `[1,4,6,7]` |
| 2 | $\left[\frac{3}{4}, 0, -z\right]$ | `[2,3,5,8]` |

Table 6: Wyckoff site: `2f`, site symmetry: `mm2`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4}, \frac{1}{2}, z\right]$ | `[1,4,6,7]` |
| 2 | $\left[\frac{3}{4}, \frac{1}{2}, -z\right]$ | `[2,3,5,8]` |

Table 7: Wyckoff site: `4g`, site symmetry: `.2.`

| No. | position | mapping |
|---|---|---|
| 1 | $[0, y, 0]$ | `[1,3]` |
| 2 | $\left[\frac{1}{2}, -y, 0\right]$ | `[2,4]` |

*continued ...*

Table 7

| No. | position | mapping |
|---|---|---|
| 3 | $[0,\ -y,\ 0]$ | `[5,7]` |
| 4 | $\left[\frac{1}{2},\ y,\ 0\right]$ | `[6,8]` |

Table 8: Wyckoff site: `4h`, site symmetry: `.2.`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0,\ y,\ \frac{1}{2}\right]$ | `[1,3]` |
| 2 | $\left[\frac{1}{2},\ -y,\ \frac{1}{2}\right]$ | `[2,4]` |
| 3 | $\left[0,\ -y,\ \frac{1}{2}\right]$ | `[5,7]` |
| 4 | $\left[\frac{1}{2},\ y,\ \frac{1}{2}\right]$ | `[6,8]` |

Table 9: Wyckoff site: `4i`, site symmetry: `.m.`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ 0,\ z]$ | `[1,7]` |
| 2 | $\left[x+\frac{1}{2},\ 0,\ -z\right]$ | `[2,8]` |
| 3 | $[-x,\ 0,\ -z]$ | `[3,5]` |
| 4 | $\left[\frac{1}{2}-x,\ 0,\ z\right]$ | `[4,6]` |

Table 10: Wyckoff site: `4j`, site symmetry: `.m.`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\ \frac{1}{2},\ z\right]$ | `[1,7]` |
| 2 | $\left[x+\frac{1}{2},\ \frac{1}{2},\ -z\right]$ | `[2,8]` |
| 3 | $\left[-x,\ \frac{1}{2},\ -z\right]$ | `[3,5]` |
| 4 | $\left[\frac{1}{2}-x,\ \frac{1}{2},\ z\right]$ | `[4,6]` |

Table 11: Wyckoff site: `4k`, site symmetry: `m..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4},\ y,\ z\right]$ | `[1,6]` |
| 2 | $\left[\frac{3}{4},\ -y,\ -z\right]$ | `[2,5]` |
| 3 | $\left[\frac{3}{4},\ y,\ -z\right]$ | `[3,8]` |
| 4 | $\left[\frac{1}{4},\ -y,\ z\right]$ | `[4,7]` |

Table 12: Wyckoff site: `8l`, site symmetry: `1`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ y,\ z]$ | `[1]` |
| 2 | $\left[x+\frac{1}{2},\ -y,\ -z\right]$ | `[2]` |
| 3 | $[-x,\ y,\ -z]$ | `[3]` |

*continued ...*

Table 12

| No. | position | mapping |
|---|---|---|
| 4 | $\left[\frac{1}{2} - x, -y, z\right]$ | [4] |
| 5 | $[-x, -y, -z]$ | [5] |
| 6 | $\left[\frac{1}{2} - x, y, z\right]$ | [6] |
| 7 | $[x, -y, z]$ | [7] |
| 8 | $\left[x + \frac{1}{2}, y, -z\right]$ | [8] |